IMDB MOVIE ANALYSIS

1. Project Overview

This project, The IMDB MovieAnalysis, provides a robust, data-driven analysis of cinematic success.The entertainment industry is a complex, multi-billion dollar sector where success is often attributed to a blend of creative vision and strategic decision-making. By applying data preprocessing techniques using the Python Pandas library and subsequent advanced visualization in Tableau, this analysis aims to move beyond anecdotal evidence to provide actionable, data-driven insights.

2. Tools and Technologies Used

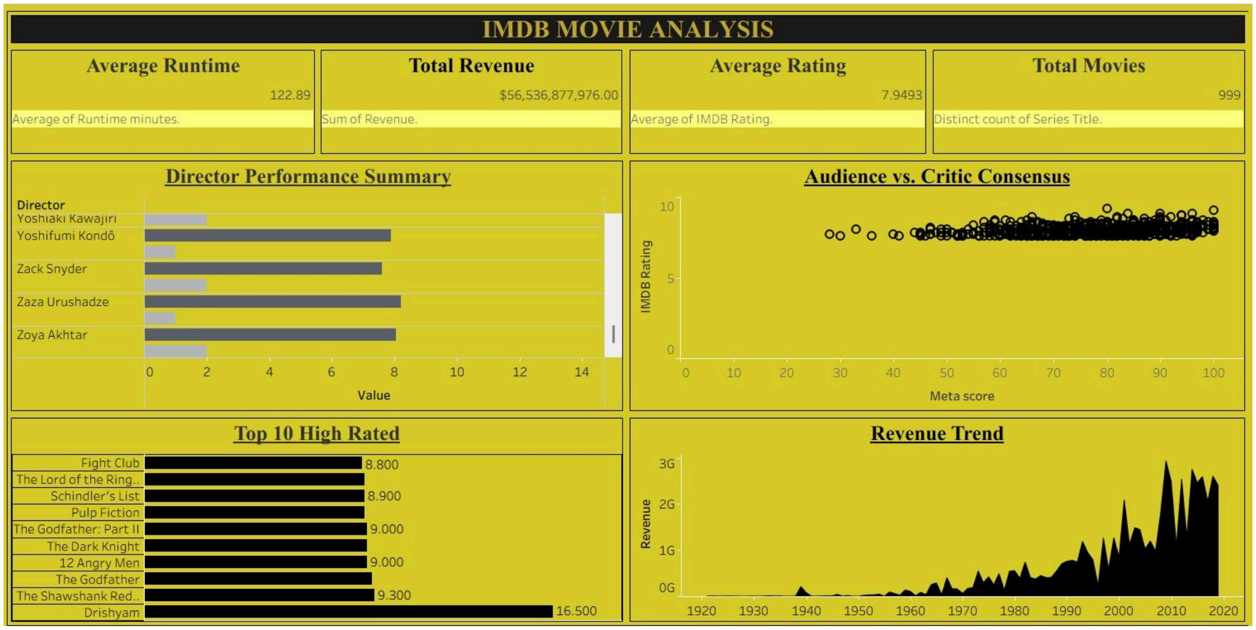
Category	**Tools/Technologies**
Data Pre processing	Python (Google Collab)
Visualization	Tableau
Data Source	IMDB movies(csv file)

3. Data Pre processing & Featuring

```python
#1. Import Liabararies and read data file
import pandas as pd
import numpy as np

df=pd.read_csv('imdb_movies.csv')

#2. Rename column 'Gross' and remove commas and convert to float
df.rename(columns={'Gross': 'Revenue'}, inplace=True)
df['Revenue'] = df['Revenue'].astype(str).str.replace(',', '', regex=False)
df['Revenue'] = pd.to_numeric(df['Revenue'], errors='coerce')

#3. Clean 'Runtime' and remove ' min' and convert to integer.
df.rename(columns={'Runtime': 'Runtime_str'}, inplace=True)
df['Runtime_minutes'] = df['Runtime_str'].str.replace(' min', '', regex=False)
df['Runtime_minutes'] = pd.to_numeric(df['Runtime_minutes'], errors='coerce').astype('Int64')

#5. Clean 'Released_Year'and convert to integer, coercing errors (if any non-numeric year exists).
df['Released_Year'] = pd.to_numeric(df['Released_Year'], errors='coerce').astype('Int64')

# 6. Final Data Selection
columns_to_keep = [
    'Series_Title', 'Released_Year', 'Runtime_minutes', 'Genre',
    'IMDB_Rating', 'Meta_score', 'Director', 'Star1', 'No_of_Votes',
    'Revenue'
]
df_clean = df[columns_to_keep].copy()

#7. Save the cleaned data
df_clean.to_csv('clean_imdb_data.csv', index=False)
```

4. Tableau Visualization

- Following the data preprocessing using Python, the cleaned data file was imported into Tableau.
- The first step was to establish the Key Performance Indicators (KPIs) to provide an immediate summary of the dataset's scale and quality. Each KPI (Total revenue, Total movies, Average runtime, Average rating) was created on a separate sheet, utilizing the Text Mark card.
- Various graphical representations are created in different sheets that focus on the temporal and foundational metrics of the dataset, forming the primary content area of the dashboard such as Revenue trend, top 10 highest rated movies,score disparity,director performance etc.
- The final dashboard was assembled in a hierarchical layout using Horizontal Containers



5. Insights & Findings

The analysis of the IMDB Top 1000 dataset reveals that peak financial performance is concentrated in the 2000s and 2010s, primarily driven by Action and Adventure films, yet the highest-grossing movies typically fall in the 8.0 to 8.5 IMDB Rating range, underscoring the importance of mass appeal over a perfect critical score. The data shows a strong overall consensus between critic scores and audience ratings, though the scatter plot highlights specific "Audience Favorites" where public opinion significantly exceeds critic reviews. Furthermore, the dataset establishes an average runtime of 120 minutes for top-tier films and confirms the exceptional consistency of certain directors, such as Christopher Nolan, in maintaining high average IMDB ratings, while modern films dominate the vote count, reflecting increased global audience engagement.

6. Conclusion

The IMDB Movie Analysis Dashboard successfully provides a quantitative framework for understanding cinematic success by linking financial metrics to critical and audience reception.

Key Conclusion:

- Critical Success (9.0+ Rating) is rare and often driven by dramatic quality.

- Commercial Success (Peak Revenue) is more common and driven by the mass appeal of the Action/Adventure genres in the high-8.0 rating range.

The project demonstrates that while quality is important (as all films are highly rated), strategic genre choice and a reliable director are the strongest predictors of consistent high performance. The final, interactive Tableau dashboard serves as a highly effective tool for talent acquisition and content strategy in the entertainment sector.

Project Submitted By
Mariya S Akkara